MATHEMATICAL NOTES

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Contents

See 随机过程基础 Basic Stochastic Processes (Zdzisław Brzeźniak, Tomasz Zastawniak) (Z-Library).

Bachelier assumed the relative infinitesimal price increments $dX(t)$ are proportional to the increments $dW(t)$ of the Wiener process,

$$\frac{dX(t)}{X(t)} = \sigma dW(t) \iff X(t) = X(0) + \sigma \int_0^t X(t)\,\mathrm{d}W(t)$$

where σ is a positive constant, the right hand side is called the Ito integral. The solution is expected to be $xe^{W(t)}$, but in fact

$$X(t) = xe^{W(t)}e^{-\frac{t}{2}}$$

due to the non-differentiability of the paths of the Wiener process.

We define Ito integral from the same spirit of Riemann integral, but the convergence is in L^2 instead of $\mathbb{R}$, and Riemann integral

$$\sum_{i=1}^{n} f(s_i)(x_i - x_{i-1})$$

does not depend on $s_i \in [x_{i-1}, x_i]$, but in stochastic case,

$$\sum_{i=1}^{n} f(s_i)(W(x_i) - W(x_{i-1}))$$

it does.

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Exercise 7.1

Let $0 = t_0^n < t_1^n < \cdots < t_n^n = T$, where $t_j^n = \frac{jT}{n}$, be a partition of the interval $[0, T]$ into n equal parts. Find the following limits in L^2:

$$\lim_{n\to\infty} \sum_{j=0}^{n-1} W(t_j^n)\left(W(t_{j+1}^n) - W(t_j^n)\right)$$

and

$$\lim_{n\to\infty} \sum_{j=0}^{n-1} W(t_{j+1}^n)\left(W(t_{j+1}^n) - W(t_j^n)\right).$$

Hint Apply Exercise 6.29. You will need to transform the sums to make this possible. The identities

$$\begin{aligned} a(b-a) &= \frac{1}{2}\left(b^2 - a^2\right) - \frac{1}{2}(a-b)^2, \\ b(b-a) &= \frac{1}{2}\left(b^2 - a^2\right) + \frac{1}{2}(a-b)^2 \end{aligned}$$

may be of help.

Proof.

$$\begin{aligned} \sum W(t_j^n)(W(t_{j+1}^n) - W(t_j^n)) &= \sum \frac{1}{2}(W(t_{j+1}^n)^2 - W(t_j^n)^2) - \frac{1}{2}\underbrace{(W(t_j^n) - W(t_{j+1}^n))^2}_{=(\Delta_i^n W)^2} \\ &= \frac{1}{2}W(t_n^n)^2 - \frac{1}{2}W(t_0^n)^2 - \frac{1}{2}\sum(\Delta_i^n W)^2 \\ &= \frac{1}{2}(W(T)^2 - W(0)^2) - \frac{1}{2}\sum(\Delta_i^n W)^2 \\ &\to \frac{1}{2}(W(T)^2 - W(0)^2) - \frac{1}{2}T \end{aligned}$$

$$\begin{aligned} \sum W(t_{j+1}^n)(W(t_{j+1}^n) - W(t_j^n)) &= \sum \frac{1}{2}(W(t_{j+1}^n)^2 - W(t_j^n)^2) + \frac{1}{2}\underbrace{(W(t_j^n) - W(t_{j+1}^n))^2}_{=(\Delta_i^n W)^2} \\ &\to \frac{1}{2}(W(T)^2 - W(0)^2) + \frac{1}{2}T \end{aligned}$$

□

6.3.3 Sample Paths

Let

$$0 = t_0^n < t_1^n < \cdots < t_n^n = T,$$

where

$$t_i^n = \frac{iT}{n},$$

be a partition of the interval $[0, T]$ into n equal parts. We denote by

$$\Delta_i^n W = W(t_{i+1}^n) - W(t_i^n)$$

the corresponding increments of the Wiener process $W(t)$.

Exercise 6.29

Show that

$$\lim_{n\to\infty} \sum_{i=0}^{n-1} (\Delta_i^n W)^2 = T \quad \text{in } L^2.$$

Hint You need to show that

$$\lim_{n\to\infty} E\left(\left[\sum_{i=0}^{n-1} (\Delta_i^n W)^2 - T\right]^2\right) = 0.$$

Use the independence of increments to simplify the expectation. What are the expectations of $\Delta_i^n W$, $(\Delta_i^n W)^2$ and $(\Delta_i^n W)^4$?

Solution 6.29

Since the increments $\Delta_i^n W$ are independent and

$$E(\Delta_i^n W) = 0, \quad E\left((\Delta_i^n W)^2\right) = \frac{T}{n}, \quad E\left((\Delta_i^n W)^4\right) = \frac{3T^2}{n^2},$$

it follows that

$$\begin{aligned}
E\left(\left[\sum_{i=0}^{n-1} (\Delta_i^n W)^2 - T\right]^2\right) &= E\left(\left[\sum_{i=0}^{n-1} \left((\Delta_i^n W)^2 - \frac{T}{n}\right)\right]^2\right) \\
&= \sum_{i=0}^{n-1} E\left[\left((\Delta_i^n W)^2 - \frac{T}{n}\right)^2\right] \\
&= \sum_{i=0}^{n-1} \left[E\left((\Delta_i^n W)^4\right) - \frac{2T}{n} E\left((\Delta_i^n W)^2\right) + \frac{T^2}{n^2}\right] \\
&= \sum_{i=0}^{n-1} \left[\frac{3T^2}{n^2} - \frac{2T^2}{n^2} + \frac{T^2}{n^2}\right] = \frac{2T^2}{n} \to 0
\end{aligned}$$

as $n \to \infty$.

The choice of s_j makes difference. To ensure that the process adapted to $\mathcal{F}_t$, we have to take $s_j = t_j$.

Definition 7.1

We shall call $f(t), t \geq 0$ a ***random step process*** if there is a finite sequence of numbers $0 = t_0 < t_1 < \ldots < t_n$ and square integrable random variables

$\eta_0, \eta_1, \ldots, \eta_{n-1}$ such that

$$f(t) = \sum_{j=0}^{n-1} \eta_j 1_{[t_j, t_{j+1})}(t), \tag{7.2}$$

where η_j is $\mathcal{F}_{t_j}$-measurable for $j = 0, 1, \ldots, n-1$. The set of random step processes will be denoted by M^2_{step}.

Definition 7.2

The *stochastic integral* of a random step process $f \in M^2_{\text{step}}$ of the form (7.2) is defined by

$$I(f) = \sum_{j=0}^{n-1} \eta_j \left(W(t_{j+1}) - W(t_j)\right). \tag{7.3}$$

Proposition 7.1

For any random step process $f \in M^2_{\text{step}}$ the stochastic integral $I(f)$ is a square integrable random variable, i.e. $I(f) \in L^2$, such that

$$E\left(|I(f)|^2\right) = E\left(\int_0^\infty |f(t)|^2 \, dt\right).$$

Proof. Expand both sides. □

Exercise 7.2

Verify that for any random step processes $f, g \in M^2_{\text{step}}$

$$E\left(I(f)I(g)\right) = E\left(\int_0^\infty f(t)g(t)\, dt\right).$$

Hint Try to adapt the proof of Proposition 7.1. Use a common partition $0 = t_0 < t_1 < \cdots < t_n$ in which to represent both f and g in the form (7.2).

Proof. Trivial. □

Exercise 7.3

Show that $I : M^2_{\text{step}} \to L^2$ is a linear map, i.e. for any $f, g \in M^2_{\text{step}}$ and any $\alpha, \beta \in \mathbb{R}$

$$I(\alpha f + \beta g) = \alpha I(f) + \beta I(g).$$

Hint As in Exercise 7.2, use a common partition $0 = t_0 < t_1 < \cdots < t_n$ in which to represent both f and g in the form (7.2).

Proof. Trivial. □

Definition 7.3

We denote by M^2 the class of stochastic processes $f(t), t \geq 0$ such that

$$E\left(\int_0^\infty |f(t)|^2 \, dt\right) < \infty$$

and there is a sequence $f_1, f_2, \ldots \in M^2_{\text{step}}$ of random step processes such that

$$\lim_{n\to\infty} E\left(\int_0^\infty |f(t) - f_n(t)|^2 \, dt\right) = 0. \tag{7.4}$$

In this case we shall say that the sequence of random step processes $f_1, f_2, \ldots$ *approximates* f in M^2.

Definition 7.4

We call $I(f) \in L^2$ the *Itô stochastic integral* (from 0 to ∞) of $f \in M^2$ if

$$\lim_{n\to\infty} E\left(|I(f) - I(f_n)|^2\right) = 0 \tag{7.5}$$

for any sequence $f_1, f_2, \ldots \in M^2_{\text{step}}$ of random step processes that approximates f in M^2, i.e. such that (7.4) is satisfied. We shall also write

$$\int_0^\infty f(t) \, dW(t)$$

in place of $I(f)$.

The Ito stochastic integral is defined by approximation.

Proposition 7.2

For any $f \in M^2$ the stochastic integral $I(f) \in L^2$ exists, is unique (as an element of L^2, i.e. to within equality a.s.) and satisfies

$$E\left(|I(f)|^2\right) = E\left(\int_0^\infty |f(t)|^2\, dt\right). \tag{7.6}$$

Proof. Prove by approximation in L^2 and M^2. □

Exercise 7.4

Show that for any $f, g \in M^2$

$$E\left(I(f)I(g)\right) = E\left(\int_0^\infty f(t)g(t)\, dt\right).$$

Hint **Write the left-hand side in terms of $E\left(|I(f)+I(g)|^2\right)$ and $E\left(|I(f)-I(g)|^2\right)$, the right-hand side in terms of $E\left(\int_0^\infty |f(t)+g(t)|^2\, dt\right)$ and $E\left(\int_0^\infty |f(t)-g(t)|^2\, dt\right)$ and then use (7.6).**

We define the Ito integral over $[0, T]$.

Definition 7.5

For any $T > 0$ we shall denote by M_T^2 the space of all stochastic processes $f(t), t \geq 0$ such that

$$1_{[0,T)} f \in M^2$$

The Itô stochastic integral (from 0 to T) of $f \in M_T^2$ is defined by

$$I_T(f) = I\left(1_{[0,T)} f\right). \tag{7.7}$$

We shall also write

$$\int_0^T f(t)\, dW(t)$$

in place of $I_T(f)$.

Exercise 7.5

Show that each random step process $f \in M^2_{\text{step}}$ belongs to M^2_t for any $t > 0$ and

$$I_t(f) = \int_0^t f(s)\, dW(s)$$

is a martingale.

Hint The stochastic integral of a random step process f is given by the sum (7.3). What is the conditional expectation of the jth term of this sum given $\mathcal{F}_s$ if $s < t_j$? What is it when $s \geq t_j$?

Solution 7.5

If $f \in M^2_{\text{step}}$ is a random step process, then so is $1_{[0,t]}f \in M^2_{\text{step}} \subset M^2$ for any $t > 0$. This in turn implies that $f \in M^2_t$ for any $t > 0$.

We shall verify that $I_t(f)$ is a martingale with respect to the filtration $\mathcal{F}_t$. Let $0 \leq s < t$ and suppose that $f \in M^2_{\text{step}}$ can be written in the form (7.2), where

$$0 = t_0 < t_1 < \cdots < t_k = s < t_{k+1} < \cdots < t_m = t < t_{m+1} < \cdots < t_n.$$

Such a partition $t_0, \ldots, t_n$ can always be obtained by adding the points s and t if necessary. We shall denote the increment $W(t_{j+1}) - W(t_j)$ by $\Delta_j W$ as in the proof of Proposition 7.1. Then

$$1_{[0,t]} f = \sum_{j=0}^{m-1} \eta_j 1_{[t_j, t_{j+1}]}$$

and

$$I_t(f) = I(1_{[0,t]} f) = \sum_{j=0}^{m-1} \eta_j \Delta_j W,$$

which is adapted to $\mathcal{F}_t$ and square integrable, and so integrable. It remains to compute

$$E\left(I_t(f)|\mathcal{F}_s\right) = E\left(I(1_{[0,t]} f)|\mathcal{F}_s\right) = \sum_{j=0}^{m-1} E\left(\eta_j \Delta_j W|\mathcal{F}_s\right).$$

If $j < k$, then η_j and $\Delta_j W$ are $\mathcal{F}_s$-measurable and

$$E\left(\eta_j \Delta_j W|\mathcal{F}_s\right) = \eta_j \Delta_j W.$$

If $j \geq k$, then $\mathcal{F}_s \subset \mathcal{F}_{t_j}$ and

$$\begin{aligned} E\left(\eta_j \Delta_j W|\mathcal{F}_s\right) &= E\left(E\left(\eta_j \Delta_j W|\mathcal{F}_{t_j}\right)|\mathcal{F}_s\right) \\ &= E\left(\eta_j E\left(\Delta_j W|\mathcal{F}_{t_j}\right)|\mathcal{F}_s\right) \\ &= E\left(\eta_j|\mathcal{F}_s\right) E\left(\Delta_j W\right) = 0, \end{aligned}$$

since η_j is $\mathcal{F}_{t_j}$-measurable and $\Delta_j W$ is independent of $\mathcal{F}_{t_j}$. It follows that

$$E\left(I_t(f)|\mathcal{F}_s\right) = \sum_{j=0}^{k-1} \eta_j \Delta_j W = I(1_{[0,s]} f) = I_s(f).$$

Proof. □

To verify the existence of integral, it's not always easy to find the approximation of step functions. Similar to Riemann integral, "continuous functions are always integrable", there is a corresponding argument for Ito integral.

Theorem 7.1

Let $f(t), t \geq 0$ be a stochastic process with a.s. continuous paths adapted to the filtration $\mathcal{F}_t$. Then

1) $f \in M^2$, i.e. the Itô integral $I(f)$ exists, whenever

$$E\left(\int_0^\infty |f(t)|^2 \, dt\right) < \infty; \tag{7.8}$$

2) $f \in M_T^2$, i.e. the Itô integral $I_T(f)$ exists, whenever

$$E\left(\int_0^T |f(t)|^2 \, dt\right) < \infty. \tag{7.9}$$

Proof. Omitted. □

Exercise 7.6

Show that the Wiener process $W(t)$ belongs to M_T^2 for each $T > 0$.

Hint Apply part 2) of Theorem 7.1.

Exercise 7.7

Show that $W(t)^2$ belongs to M_T^2 for each $T > 0$.

Hint Once again, apply part 2) of Theorem 7.1.

Definition 7.6

A stochastic process $f(t), t \geq 0$ is called ***progressively measurable*** if for any $t \geq 0$

$$(s, \omega) \mapsto f(s, \omega)$$

is a measurable function from $[0, t] \times \Omega$ with the σ-field $\mathcal{B}[0, t] \bar{\times} \mathcal{F}$ to $\mathbb{R}$. Here $\mathcal{B}[0, t] \bar{\times} \mathcal{F}$ is the product σ-field on $[0, t] \times \Omega$, that is, the smallest σ-field containing all sets of the form $A \times B$, where $A \subset [0, t]$ is a Borel set and $B \in \mathcal{F}$.

Theorem 7.2

1) The space M^2 consists of all progressively measurable stochastic processes $f(t), t \geq 0$ such that

$$E\left(\int_0^\infty |f(t)|^2 \, dt\right) < \infty.$$

2) The space M_T^2 consists of all progressively measurable stochastic processes $f(t), t \geq 0$ such that

$$E\left(\int_0^T |f(t)|^2 \, dt\right) < \infty.$$

Theorem 7.3

The following properties hold for any $f, g \in M_t^2$, any $\alpha, \beta \in \mathbb{R}$, and any $0 \leq s < t$:

1) *linearity*

$$\int_0^t (\alpha f(r) + \beta g(r)) \, dW(r) = \alpha \int_0^t f(r) \, dW(r) + \beta \int_0^t g(r) \, dW(r);$$

2) *isometry*

$$E\left(\left|\int_0^t f(r) \, dW(r)\right|^2\right) = E\left(\int_0^t |f(r)|^2 \, dr\right);$$

3) *martingale property*

$$E\left(\int_0^t f(r) \, dW(r) \middle| \mathcal{F}_s\right) = \int_0^s f(r) \, dW(r).$$

Lemma 7.1

If ξ and $\xi_1, \xi_2, \ldots$ are square integrable random variables such that $\xi_n \to \xi$ in L^2 as $n \to \infty$, then

$$E(\xi_n|\mathcal{G}) \to E(\xi|\mathcal{G}) \quad \text{in } L^2 \text{ as } n \to \infty$$

for any σ-field $\mathcal{G}$ on Ω contained in $\mathcal{F}$.

Proof. Use Jensen's formula. □

Definition 7.7

Let $\xi(t)$ and $\zeta(t)$ be stochastic processes defined for $t \in T$, where $T \subset \mathbb{R}$. We say that the processes are ***modifications*** (or ***versions***) of one another if

$$P\{\xi(t) = \zeta(t)\} = 1 \quad \text{for all } t \in T. \tag{7.13}$$

Next we consider the stochastic differentiation, which is different from the classical calculus for smooth function.

Theorem 7.5 (Itô formula, simplified version)

Suppose that $F(t,x)$ is a real-valued function with continuous partial derivatives $F'_t(t,x)$, $F'_x(t,x)$ and $F''_{xx}(t,x)$ for all $t \geq 0$ and $x \in \mathbb{R}$. We also assume that the process $F'_x(t, W(t))$ belongs to M_T^2 for all $T \geq 0$. Then $F(t, W(t))$ is an Itô process such that

$$\begin{aligned} F(T, W(T)) - F(0, W(0)) \;=\; & \int_0^T \left(F'_t(t, W(t)) + \frac{1}{2} F''_{xx}(t, W(t)) \right) dt \\ & + \int_0^T F'_x(t, W(t))\, dW(t) \quad \text{a.s.} \end{aligned} \tag{7.17}$$

In differential notation this formula can be written as

$$dF(t, W(t)) = \left(F'_t(t, W(t)) + \frac{1}{2} F''_{xx}(t, W(t)) \right) dt + F'_x(t, W(t))\, dW(t). \tag{7.18}$$

Remark 7.2

Compare the latter with the chain rule

$$dF(t, x(t)) = F'_t(t, x(t))\, dt + F'_x(t, x(t))\, dx(t).$$

for a smooth function $x(t)$, where $dx(t)$ is understood as a shorthand notation for $x'(t)\, dt$. The additional term $\frac{1}{2}F''_{xx}(t, W(t))\, dt$ in (7.18) is called the *Itô correction.*

Example 7.4

For $F(t,x) = x^2$ we have $F'_t(t,x) = 0$, $F'_x(t,x) = 2x$ and $F''_{xx}(t,x) = 2$. The Itô formula gives

$$d\left(W(t)^2\right) = dt + 2W(t)\, dW(t),$$

which is the same equality as in Exercise 7.8.

Example 7.5

For $F(t,x) = x^3$ we have $F'_t(t,x) = 0$, $F'_x(t,x) = 3x^2$ and $F''_{xx}(t,x) = 6x$. By the Itô formula we obtain the same equality

$$d\left(W(t)^3\right) = 3W(t)\, dt + 3W(t)^2\, dW(t)$$

as in Exercise 7.10.

Exercise 7.14 (exponential martingale)

Show that the exponential martingale $X(t) = e^{W(t)}e^{-\frac{t}{2}}$ is an Itô process and verify that it satisfies the equation

$$dX(t) = X(t)\, dW(t).$$

Hint Use the Itô formula with $F(t,x) = e^x e^{-\frac{t}{2}}$.

$$F'_t(t,x) = -\frac{1}{2}e^{x-\frac{t}{2}}, \quad F''_{xx}(t,x) = e^{x-\frac{t}{2}}, \quad F'_x(t,x) = e^{x-\frac{t}{2}}$$

are continnuous for all $t \geq 0$ and $x \in \mathbb{R}$. Assume that $F'_x(t, W(t)) \in M_T^2$ for all $T \geq 0$. Then $F(t, W(t))$ is Ito process and

$$dF(t, W(t)) = \underbrace{\left(-\frac{1}{2}e^{W(t)}e^{-\frac{t}{2}} + \frac{1}{2}e^{W(t)}e^{-\frac{t}{2}}\right)}_{=0} dt + \underbrace{e^{W(t)}e^{-\frac{t}{2}}}_{=F(t,W(t))} dW(t)$$

i.e.

$$dX(t) = X(t)dW(t)$$

Theorem 7.6 (Itô formula, general case)

Let $\xi(t)$ be an Itô process as above. Suppose that $F(t, x)$ is a real-valued function with continuous partial derivatives $F'_t(t, x)$, $F'_x(t, x)$ and $F''_{xx}(t, x)$ for all $t \geq 0$ and $x \in \mathbb{R}$. We also assume that the process $b(t)F'_x(t, \xi(t))$ belongs to M_T^2 for all $T \geq 0$. Then $F(t, \xi(t))$ is an Itô process such that

$$\begin{aligned} dF(t, \xi(t)) \;=\; & \left(F'_t(t, \xi(t)) \;+ F'_x(t, \xi(t))\, a(t) + \frac{1}{2}F''_{xx}(t, \xi(t))\, b(t)^2\right) dt \\ & + F'_x(t, \xi(t))\, b(t)\, dW(t). \end{aligned} \tag{7.20}$$

Then we consider the Stochastic Differential Equations.

$$d\xi(t) = f(\xi(t))dt + g(\xi(t))dW(t)$$

Example 7.6

The stochastic differential equation

$$dX(t) = X(t)\, dW(t) \tag{7.22}$$

was used as a motivation for developing Itô stochastic calculus at the beginning of the present chapter. In Exercise 7.14 it was verified that the exponential martingale

$$X(t) = e^{W(t)}e^{-\frac{t}{2}}$$

satisfies (7.22). It also satisfies the initial condition $X(0) = 1$. This is an example of a linear stochastic differential equation. For the solution of a general equation of this type with an arbitrary initial condition, see Exercise 7.20.

Example 7.7

In Exercise 7.16 it was shown that the Ornstein–Uhlenbeck process

$$Y(t) = \sigma e^{-\alpha t} \int_0^t e^{\alpha s}\, dW(s)$$

satisfies the stochastic differential equation

$$dY(t) = -\alpha Y(t)\, dt + \sigma\, dW(t)$$

with initial condition $Y(0) = 0$. This is an example of an inhomogeneous linear stochastic differential equation. See Exercise 7.17 for a solution with an arbitrary initial condition.

Definition 7.9

An Itô process $\xi(t)$, $t \geq 0$ is called a solution of the initial value problem

$$\begin{aligned} d\xi(t) &= f(\xi(t))\, dt + g(\xi(t))\, dW(t), \\ \xi(0) &= \xi_0 \end{aligned}$$

if ξ_0 is an $\mathcal{F}_0$-measurable random variable, the processes $f(\xi(t))$ and $g(\xi(t))$ belong, respectively, to $\mathcal{L}_T^1$ and M_T^2, and

$$\xi(T) = \xi_0 + \int_0^T f(\xi(t))\, dt + \int_0^T g(\xi(t))\, dW(t) \quad \text{a.s.} \tag{7.23}$$

for all $T \geq 0$.

We have Existence and Uniqueness theorem.

Theorem 7.7

Suppose that f and g are Lipschitz continuous functions from $\mathbb{R}$ to $\mathbb{R}$, i.e. there is a constant $C > 0$ such that for any $x, y \in \mathbb{R}$

$$\begin{aligned} |f(x) - f(y)| &\leq C\,|x - y|\,, \\ |g(x) - g(y)| &\leq C\,|x - y|\,. \end{aligned}$$

Moreover, let ξ_0 be an $\mathcal{F}_0$-measurable square integrable random variable. Then the initial value problem

$$d\xi(t) = f(\xi(t))\,dt + g(\xi(t))\,dW(t), \tag{7.24}$$

$$\xi(0) = \xi_0 \tag{7.25}$$

has a solution $\xi(t), t \geq 0$ in the class of Itô processes. The solution is unique in the sense that if $\eta(t), t \geq 0$ is another Itô process satisfying (7.24) and (7.25), then the two processes are identical a.s., that is,

$$P\left\{\xi(t) = \eta(t) \text{ for all } t \geq 0\right\} = 1.$$

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